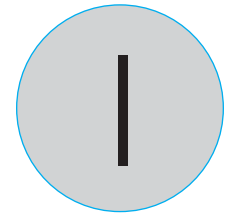


General Concepts



P A R T



C H A P T E R

Therapeutic Exercise: Foundational Concepts

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exercise affect tissues of the body and body systems and how those exercise-induced effects have an impact on key aspects of physical function. A therapist must also integrate and apply knowledge of anatomy, physiology, kinesiology, pathology, and the behavioral sciences across the continuum of patient/client management from the initial examination to discharge planning. To develop therapeutic exercise programs that culminate in positive and meaningful functional outcomes for patients and clients, a therapist must understand the relationship between physical function and disability and appreciate how the application of the process of disablement to patient/client management facilitates the provision of effective and efficient health-care services. Finally, a therapist, as a patient/client educator, must know and apply principles of motor learning and motor skill acquisition to exercise instruction and functional training. Therefore, the purpose of this chapter is to present an overview of the scope of therapeutic exercise interventions used by physical therapists. We discuss models of

disablement and patient/client management as they relate to therapeutic exercise and explore strategies for teaching and progressing exercises and functional motor skills based on principles of motor learning.

THERAPEUTIC EXERCISE: IMPACT ON PHYSICAL FUNCTION

the continuum of care of patients and clients, therapeutic exercise takes its place as one of the key elements that lies at the center of programs designed to improve or restore an individual's function or to prevent dysfunction.²

Definition of Therapeutic Exercise

*Therapeutic exercise*² is the systematic, planned performance of bodily movements, postures, or physical activities intended to provide a patient/client with the means to

- Remediate or prevent impairments
- Improve, restore, or enhance physical function
- Prevent or reduce health-related risk factors
- Optimize overall health status, fitness, or sense of well-being

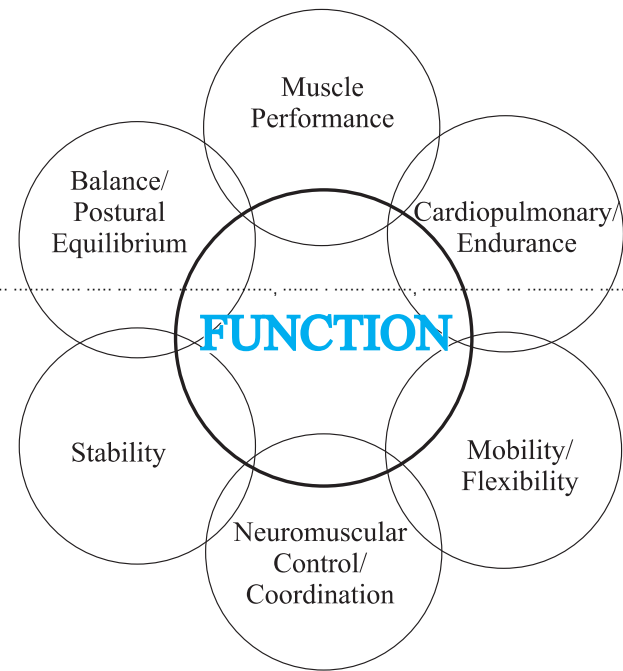


FIGURE 1.1 Interrelated aspects of physical function.

to maintain or move the body (center of mass) within the available base of support without falling; the ability to move the body in equilibrium with gravity via interaction of the sensory and motor systems.^{2,66,74,88,124,127,128}

Cardiopulmonary fitness. The ability to perform low-intensity, repetitive, total body movements (walking, jogging, cycling, swimming) over an extended period of time^{1,81}; a synonymous term is *cardiopulmonary endurance*.

cle firing combined with the appropriate intensity of muscular contraction leading to the effective initiation, guiding, and grading of movement. It is the basis of smooth, accurate, efficient movement and occurs at a conscious or automatic level.^{123,127}

Flexibility. The ability to move freely, without restriction; used interchangeably with mobility.

to move or be moved in order to allow the occurrence of range of motion (ROM) for functional activities (functional ROM).^{2,134} Passive mobility is dependent on soft tissue (contractile and noncontractile) extensibility; in addition, active mobility requires neuromuscular activation.

tension and do physical work. Muscle performance encompasses strength, power, and muscular endurance.²

Neuromuscular control. Interaction of the sensory and motor systems that enables synergists, agonists and antagonists, as well as stabilizers and neutralizers to anticipate or

respond to proprioceptive and kinesthetic information and, subsequently, to work in correct sequence to create coordinated movement.⁷²

Postural control, postural stability, and equilibrium.

Used interchangeably with static or dynamic balance.^{50,124,127}

The systems of the body that control each of these aspects of physical function react, adapt, and develop in response to forces and physical stresses (stress = force/area) placed upon tissues that make up body systems.^{81,84} Gravity, for example, is a constant force that affects the musculoskeletal, neuromuscular, and circulatory systems. Additional forces, incurred during routine physical activities, help the body maintain a functional level of strength, cardiopulmonary fitness, and mobility. Imposed forces and physical stresses that are excessive can cause acute injuries, such as sprains and fractures, or chronic conditions, such as repetitive stress disorders.⁸⁴ The absence of typical forces on the body can also cause degeneration, degradation, or deformity. For example, the absence of normal weight bearing associated with prolonged bed rest or immobilization weakens muscle and bone.^{8,20,69,84,108} Prolonged inactivity also leads to decreased efficiency of the circulatory and pulmonary systems.¹

Impairment of any one or more of the body systems and subsequent impairment of any aspect of physical function, separately or jointly, can result in functional limitation and disability. Therapeutic exercise interventions involve the application of carefully graded physical stresses and forces that are imposed on impaired body systems, specific tissues, or individual structures in a controlled, progressive, safely executed manner to reduce physical impairments and improve function.

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NOTE: Although joint mobilization techniques are often classified as manual therapy procedures, not therapeutic exercise,² the authors of this textbook have chosen to include joint mobilization procedures under the broad

BOX 1.1 Therapeutic Exercise Interventions

- Aerobic conditioning and reconditioning
- Muscle performance exercises: strength, power, and endurance training
- Stretching techniques including muscle-lengthening procedures and joint mobilization techniques.....^{50,127,134}
- Neuromuscular control, inhibition, and facilitation techniques and posture awareness training
- Postural control, body mechanics, and stabilization exercises
- Balance exercises and agility training
- Relaxation exercises
- Breathing exercises and ventilatory muscle training
- Task-specific functional training

scope of soft tissue stretching techniques.

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also affects patient safety. Adequate space and a proper support surface for exercise are necessary prerequisites for patient safety. If exercise equipment is used in the clinical setting or at home, to ensure patient safety the equipment must be well maintained and in good working condition, must fit the patient, and must be applied and used properly.

with which a patient performs an exercise affects safety, including proper posture or alignment of the body, execution of the correct movement patterns, and performing each exercise with the appropriate intensity, speed, and duration. A patient must be informed of the signs of

fatigue, the relationship of fatigue to the risk of injury, and the importance of rest for recovery during and after an exercise routine. When a patient is being directly supervised in a clinical or home setting while learning an exercise program, the therapist can control these variables. However, when a patient is carrying out an exercise program independently at home or at a community fitness facility, patient safety is enhanced and the risk of injury or re-injury is minimized by effective exercise instruction and patient education. Suggestions for effective exercise instruction and patient education are discussed in a later section of this chapter.

precautions, contraindications, and safety considerations are addressed for the management of specific pathologies, impairments, and functional limitations and for the use and progression of specific therapeutic exercise interventions.



PROCESS AND MODELS OF DISABLEMENT

defined by a body of knowledge and clinical applications that are directed toward the elimination or resolution of disability.¹¹⁰ Understanding the disabling consequences of disease, injury, and abnormalities of development and how the risk of potential disability can be reduced, therefore, must be fundamental to the provision of effective care and services, which are geared to the restoration of meaningful function for patients and their families, significant others, and caregivers.

Disablement is a term that refers to the impact(s) and functional consequences of acute or chronic conditions, such as disease, injury, and congenital or developmental abnormalities, on specific body systems that compromise basic human performance and an individual's ability to meet necessary, customary, expected, and desired societal functions and roles.^{60,86,143} Physical therapists most commonly provide care and services to people with *physical* disability. Social, emotional, and cognitive disablement can affect physical function and vice versa and, therefore, should not be disregarded or dismissed.^{58,60}

Implications in Health Care

Knowledge of the process of disablement provides a foundation for health-care professionals to develop an

appreciation of the complex relationships among function, disability, and health.^{62,132} This knowledge, in turn, provides a theoretical framework upon which practice can be organized and research can be based, thus facilitating effective management and care of patients that is reflected by meaningful functional outcomes.^{40,62}

edge of the disablement process in health-care delivery is an understanding that the process is *not unidirectional*; that is, it is not necessarily unpreventable or irreversible.¹⁴ Furthermore, it is assumed that in most instances, depending on factors such as the severity and duration of the pathological condition, a patient's access to quality health care as well as the motivation and desires of the patient, the progression of the process can indeed be altered and the patient's function improved.^{2,14,132,133}

An understanding and application of the disablement process shifts the focus of patient management from strict treatment of a disease or injury to treatment of the *impact* that a disease, injury, or disorder has on a patient's *function* as well as the identification of the underlying causes of the patient's dysfunction. This perspective puts the person, not solely the disease or disorder, at the center of efforts to prevent or halt the progression of disablement by employing interventions that improve a patient's functional abilities while simultaneously reducing or eliminating the causes of disability.^{39,132,133}

Models of Disablement

40 Nagi86,87 110

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The conceptual frameworks of the Nagi, ICIDH, and NCMRR models of disablement, although applied widely

TABLE 1.1 Comparison of Terminology of Three Disablement Models

Model	Tissue/Cellular Level	Organ/System Level	Personal Level	Societal Level
	ICIDH* Active pathology	ICF Impairment	Functional limitation	Disability
ICIDH*	Disease	Impairment	Disability	Handicap
ICF†		Impairment of body structure/function	Activity limitation	Participation restriction

*International Classification of Impairments, Disabilities, and Handicaps.
†International Classification of Functioning, Disability, and Health.

in clinical practice and research in many health-care professions, have been criticized for their perceived focus on disease and a medical-biological view of disability as well as their lack of attention to the person with a disability.²¹ In response to these criticisms, the WHO undertook a broad revision of its conceptual framework and system for classifying disability described in its ICIDH model. Through a comprehensive consensus process over a number of years, the WHO developed the International Classification of Functioning, Disability, and Health (ICF).^{52,131-133} This new conceptual model integrates functioning and disability and is characterized as a bio-psycho-social model of disablement that provides a coherent perspective of various aspects of health. The revised model was also designed to place less emphasis on disease and greater emphasis on how people affected by health conditions live.^{21,52,132,133} The ICF model consists of the following components of health and health-related influences.

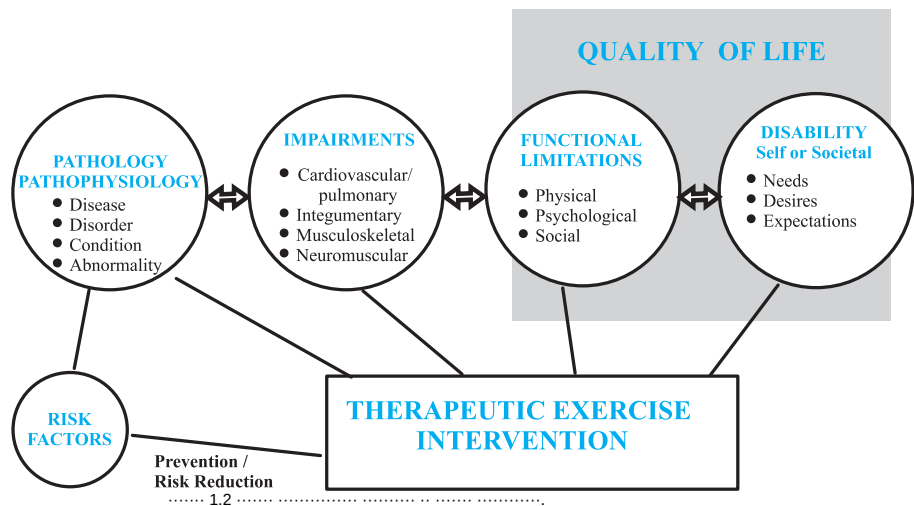
- (physiological)
- Activity limitation
- Participation restriction
- Impact of contextual factors (environmental and personal) on functioning, disability, and health

The components of the ICF model are compared in Table 1.1 to the Nagi and ICIDH models of disablement.

Use of Disablement Models and Classifications in Physical Therapy

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Figure 1.2, which was configured for this textbook, depicts a model of disablement and the potential impacts of therapeutic exercise interventions on the disablement process. The impact of risk factors has also been included in this depiction of the process. Incorporating risk factors into the



NOTE: ICF .52

Impairments are the *consequences* of pathological conditions; that is, they are the signs and symptoms that reflect abnormalities at the body system, organ, or tissue level.^{2,36,58}

Types of Impairment

Impairments can be categorized as arising from *anatomical*, *physiological*, or *psychological* alterations as well as losses or abnormalities of *structure* or *function* of a body system. Physical therapists typically provide care and services to patients with impairments that affect the following systems.

- ☐
- ☐ Neuromuscular
- ☐ Cardiovascular/pulmonary
- ☐ Integumentary

Most impairments of these body systems primarily are the result of abnormalities of physiological function or anatomical structure. Some representative examples of physical impairments commonly identified by physical therapists and managed with therapeutic exercise interventions are noted in Box 1.2.



A



B

..... 1.3 () (.....), () (.....),

tendinitis of the rotator cuff (pathology) leading to (B) limited range of shoulder elevation (an impairment) are identified during the examination.

BOX 1.2 Common Physical Impairments Managed with Therapeutic Exercise

Musculoskeletal

- Pain
- Muscle weakness/reduced torque production
- Decreased muscular endurance
- Limited range of motion due to
 - Restriction of the joint capsule
 - Restriction of periarticular connective tissue
 - Decreased muscle length
- Joint hypermobility
- Faulty posture
- Muscle length/strength imbalances

- Pain
- Impaired balance, postural stability, or control
- Incoordination, faulty timing
- Delayed motor development
- Abnormal tone (hypotonia, hypertonia, dystonia)
- Ineffective/inefficient functional movement strategies

Cardiovascular/Pulmonary

- Decreased aerobic capacity (cardiopulmonary endurance)
- Impaired circulation (lymphatic, venous, arterial)
- Pain with sustained physical activity (intermittent claudication)

Integumentary

- Skin hypomobility (e.g., immobile or adherent scarring)

tations may be *physical, social, or psychological* in nature. The focus of physical therapy interventions is on the management of limitations of physical functioning while respecting the needs of the whole person and recognizing that social and psychological influences can also limit a person's ability to function. In addition, the focus must be on those functional limitations that are most important to the patient and those that are or could be directly causing disability. When impairments cause functional limitations, a person's *quality of life* may begin to deteriorate (see Fig. 1.2). It should also be noted that a single or even several mild impairments often do not cause loss of function. Evidence suggests that the severity and complexity of impairments must reach a critical level, which is different for each person, before degradation of function begins to occur.^{95,103}

Types of Functional Limitations

Functional limitations in the physical domain deal with the performance of sensorimotor tasks, that is, total body actions that are typically *components or elements* of functional activities.^{2,89} These activities include basic activities of daily living (ADL), such as bathing, dressing, or feeding, and the more complex tasks known as instrumental activities of daily living (IADL), such as occupational tasks, school-related skills, housekeeping, and recreational activities, or community mobility (driving, using public transportation), just to name a few.

can arise from physical impairments, involve *whole-body movements*, and are necessary component motions of simple to complex daily living skills. Defining functional limi-



FIGURE 1.4 Limited ability to reach overhead (functional limitation) as the result of impaired shoulder mobility may lead to difficulty performing personal grooming or household tasks independently (disability).

1.3

Common Functional Limitations Related to Physical Tasks

Limitation of

- Reaching and grasping
- Lifting and carrying
- Pushing and pulling
- Bending and stooping
- Turning and twisting
- Throwing and catching
- Rolling
- Standing
- Squatting and kneeling
- Standing up and sitting down
- Getting in and out of bed
- Crawling, walking, running
- Ascending and descending stairs
- Hopping and jumping
- Kicking